

B.Tech. Degree V Semester Examination November 2014

ME 1502 MECHANICS OF MACHINERY

(2012 Scheme)

Time : 3 Hours

Maximum Marks : 100

PART A (Answer *ALL* questions)

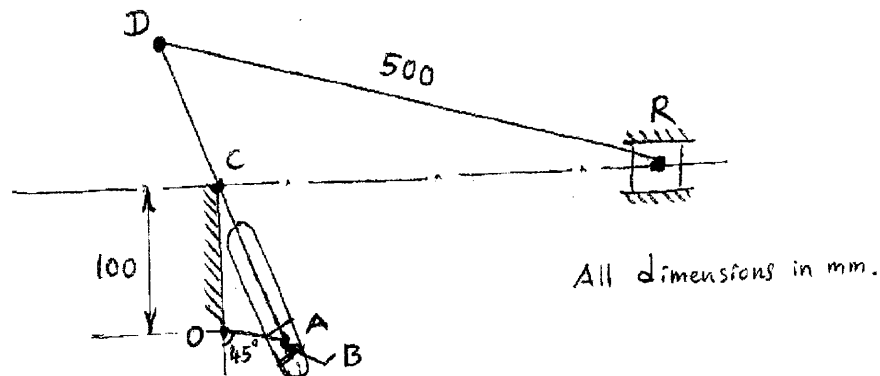
(8 x 5 = 40)

- I. (a) Define kinematic chain and mechanism.
 (b) Sketch and explain inversions of four-bar mechanism.
 (c) Explain interference and how is this avoided.
 (d) Explain the working of an epicyclic gear train with bevel wheels.
 (e) What is a cam? Classify cams and followers.
 (f) What is a screw jack? Explain the principle, on which it works.
 (g) Describe with a neat sketch the torsion dynamometer.
 (h) Briefly explain synthesis of mechanism with examples.

PART B

(4 x 15 = 60)

- II. In a Whitworth quick return motion shown in fig. OA is a crank rotating at 30 r.p.m. (15)
 in a clockwise direction. The dimensions of various links are: OA = 150mm, OC = 100mm, CD = 125mm and DR = 500mm. Determine the acceleration of the sliding block R and the angular acceleration of the slotted lever CA.



OR

(P.T.O.)

- III. (a) Explain different methods to locate instantaneous centres in a mechanism. (5)
 (b) State and prove Kennedy's theorem. (10)

- IV. (a) State law of gearing. (5)
 (b) A pair of spur wheels with 14 and 21 teeth are of involute profile and pressure angle 16° . Find maximum addenda on the pinion and gear wheel to avoid interference if module is 6mm. Also find the maximum velocity of sliding on either side of the pitch point if pinion runs at 300 rpm. (10)

OR

- V. In an epicyclic gear train the arm carries two gears A and B having 40 and 60 teeth respectively. If the arm rotates at 150 rpm in the anticlockwise direction about the centre of the gear A which is fixed, determine the speed of gear B. If the gear A instead of being fixed makes 300 rpm in the clockwise direction, what will be the speed of gear B? (15)

- VI. (a) Explain overhauling and self locking of screws. (7)
 (b) The pitch of 50mm mean diameter threaded screw of a screw jack is 12.5mm. The coefficient of friction between the screw and the nut is 0.13. Determine the torque required on the screw to raise a load of 25KN, assuming the load to rotate with the screw. Determine the ratio of the torque required to raise the load to the torque required to lower the load and also the efficiency of the machine. (8)

OR

- VII. A cam drives a flat reciprocating follower in the following manner. During first 120° rotation of the cam, follower moves outwards through a distance of 20mm with S.H.M. The follower dwells during next 30° of cam rotation. During next 120° of cam rotation, the follower moves inwards with S.H.M. The follower dwells for the next 90° cam rotation. The minimum radius of the cam is 25mm. Draw the profile of the cam. (15)

- VIII. (a) A car engine has its rated output of 10KW. Maximum torque developed is 100N-M. The clutch used is of single plate type, having two active surfaces. Axial pressure is not to exceed 0.85 bar. External diameter of the friction plate is 1.25 times the internal diameter. Determine the dimension of the friction plate and the axial force exerted by the springs. Assume uniform wear and coefficient of friction is 0.3. (7)
 (b) A multiple disc clutch has five plates having four active parts of frictional surfaces. Determine the maximum axial intensity of pressure between the discs for transmitting 18KW at 500 rpm. If the outer and inner radii of the friction surface are 12.5cm and 7.5 cm respectively. Assume uniform wear and the coefficient of friction as 3. (8)

OR

- IX. Obtain Freudenstein's equation. Illustrate how Freudenstein's equation can be used in the synthesis of a four bar linkage. (15)
